



Does repair matter?

Unobserved decisions in a dynamic discrete choice model

Rodrigo Antonio Aldrette Salas

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Objectives

- Propose a model of Dynamic Discrete Choice for a secondary market that takes into account repair and maintenance.
- Propose an identification strategy for markets where repair and maintenance are crucial.
- Measure the effect of repair/maintenance prices on secondary market prices, lifetime of durable goods and heterogeneous welfare effects as per type of consumer.

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Importance

- Repair and maintenance prices determine the usable life of durable goods.
- Durable goods constitute the main wealth of households
- The pandemic shock on repair prices could have been an important channel for the shock on the wealth on households.
- Policy regarding repair and maintenance prices can have important environmental effects.

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Mexico's context

Los costes de reparación del automóvil han subido casi un 20% en el último año. Estas son las razones

La mejora de la tecnología, el aumento de los costes laborales, el incremento de los accidentes y los continuos problemas con las cadenas de suministro de piezas han encarecido la reparación de autos.

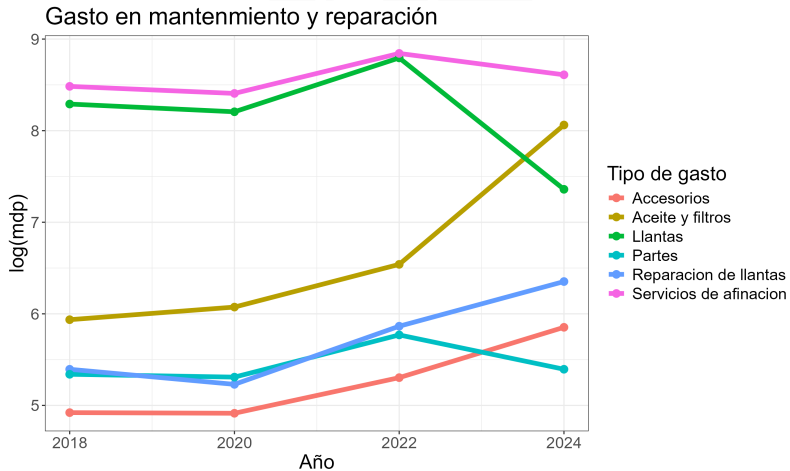
La Fed elevó de nuevo las tasas de interés activas llevándolas a su punto más alto en 22 años

En cuanto a precio, **el gasto por visita en el distribuidor es más alto, con 3 mil 463 pesos por visita; mientras que en los talleres independientes el ticket promedio del servicio es de 2 mil 432 pesos.**

"El negocio de servicios se vio muy afectado por la pandemia, pero tanto los distribuidores como los talleres independientes han visto un incremento tanto en las visitas como en los ingresos por visita", dijo Gerardo Gómez, director general de J.D. Power de México.



Mexico's context





The data

First we begin to describe the data at hand so the model is put into more context. It consists three databases spanning 13 years from Denmark. The databases are

- **Counts** Cell-based data of trade of automobiles
- **Demo** The demography of the types of consumer
- **Counts scrap** The frequency of scrapped vehicles
- **Car attributes** The attributes of the cars considered

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Counts database

This one is at the year (t), consumer type (τ), model (j) and age of the car (a) level. The observations describe what action was taken (choosing some car or no car) and the odometer from that car that they acquired (average).

tau	s_car_type	s_car_age	d_car_type	d_car_age	year	count	d_fuel_price	d_km
1	-1	-1	-1	-1	1996	193719		
1	-1	-1	1	0	1996	2072	8.064856309	37.5976122
1	-1	-1	1	1	1996	251	7.772069358	41.7049229
1	-1	-1	1	2	1996	470	7.548685246	45.1266583
1	-1	-1	1	3	1996	379	7.497405956	45.0238089
1	-1	-1	1	4	1996	616	7.3363444	46.1546132
1	-1	-1	1	5	1996	591	7.385188871	40.2615405
1	-1	-1	1	6	1996	500	7.35107329	40.3113139



Demo

The demography of the types of consumer

tau	year	count	age	bigcity1	nkids	real_inc	single	wd
1	1996	558820	53.5479627	0.221735085	0.515192728	308282.229	0	0
1	1997	560874	53.7359951	0.220323994	0.513120594	306139.642	0	0
1	1998	556487	54.0997472	0.216666337	0.505760242	306719.856	0	0
1	1999	557468	54.2750041	0.215650046	0.502497004	305531.23	0	0
1	2000	570647	54.1293926	0.214975282	0.513036956	306472.441	0	0
1	2001	554658	54.3955293	0.215419231	0.506450822	307827.073	0	0
1	2002	501088	54.8688254	0.221513986	0.486028402	315943.919	0	0
1	2003	491485	55.3376654	0.220383125	0.475788681	315791.084	0	0



Scrap count

This includes the **probability of scrap** gotten from the frequency.

tau	s_car_type	s_car_age	year	count	pr_scrap
1	-1	-1	1996	246590	
1	-1	-1	1997	218613	
1	-1	-1	1998	187052	
1	-1	-1	1999	163060	
1	-1	-1	2000	158377	
1	-1	-1	2001	152714	
1	-1	-1	2002	144254	
1	-1	-1	2003	137420	
1	-1	-1	2004	134147	



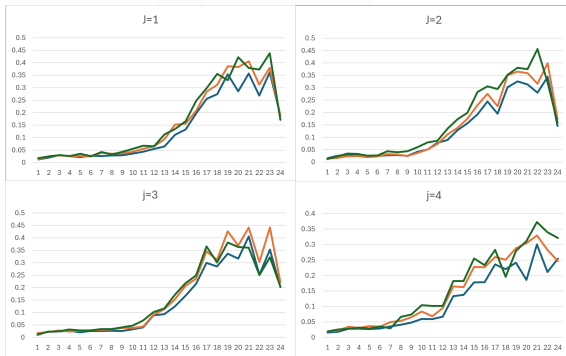
Car attributes

car_type	car_age	count	beta	diesel	fuel_eff	price_new	total_weight_tons	year
1	0	24865	0.875326844	0.001246732	13.7061679	154066.032	1.493399775	1996
1	1	21947	0.870945298	0.001230236	13.9165905	145674.992	1.457680574	1996
1	2	22987	0.871481819	0.000261017	13.5778051	147875.883	1.460712182	1996
1	3	13663	0.871965033	0.000439142	13.4106866	152257.492	1.474507624	1996
1	4	18206	0.872423094	0	13.2459056	152747.838	1.468139769	1996
1	5	16768	0.870438723	0	13.25067	147081.227	1.440703615	1996
1	6	14446	0.871168067	0	13.3807918	133781.246	1.398775871	1996
1	7	13329	0.871814306	7.50244E-05	13.2920363	127288.871	1.380044878	1996
1	8	15978	0.871467248	6.25861E-05	13.0407892	127001.514	1.378141873	1996
1	9	17591	0.871357712	0	12.7091069	128888.338	1.386442585	1996
1	10	21184	0.871033801	0	12.950262	115028.71	1.334518646	1996
1	11	22498	0.870663831	0	12.6521723	116437.58	1.338777269	1996
1	12	12532	0.869899996	7.97957E-05	12.3779782	117842.075	1.338200473	1996
1	13	10846	0.86982826	0	12.0311687	119894.203	1.348257791	1996
1	14	3581	0.869276245	0	11.8792061	118184.301	1.336133762	1996
1	15	4721	0.869616345	0	11.9623782	107598.601	1.303658123	1996
1	16	3197	0.869668644	0	11.9412058	100311.095	1.280860494	1996



Why now the data?

This model relies heavily in the variation and heterogeneity coming from scrap probability. We can observe that in the data. There are estational trends (it is mentioned in the paper). The x axis is probability of scrap and the x axis is age of car. The different lines are types of consumers.





Quick Model Overview

- We model the secondary market of cars
- The state space is described by brand, age and probability of scrap, all observed by the econometrician.
- Infinite horizon, everything stationary.
- Our equilibrium is defined by the price vector brand/age and the distribution of consumers on each state.
- Repair decisions are unobserved, but identified through the repair price thanks to Hu & Xin (2024).
- Repair/maintenance affects probability of scrap.



What have I advanced with?

This week we did a Monte Carlo analysis of a simplified Rust(1987) model so that I can have the backbone of the estimation. It was solved with Nested Fixed Point (NFXP) as my model can also be solved through maximum likelihood. Repository in my Github rojoaldrette

```
29
30 theta = [4.0, 0.09]
31 g = make_params(theta)
32 #g1 = replace(g, replacement_cost=4)
33
34
35 # Get probabilities #####
36 p1, p0 = calc_probs(g)
37
38
39 # Generate dataset #####
40
41 df = gen_dataset(g, 123)
42
43
44 # Estimation #####
45 theta=[8.0, 0.015]
46 theta0 = [1.0, 1.0]
47 results = montecarlo(theta)
48
49
50
```

A Simple Rust Model

Rodrigo Antonio Aldrette Salas

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1 Introduction

In this document we develop a simple Rust model so we can show the power of JAX when using a NFXP algorithm for estimation.

2 The Model

The model follows closely the version from Hotz (1994). In each month t , the owner decides whether or not to replace a bus's engine ($J = 2$) to minimize the discounted costs of maintaining it. Everytime he is faced with a bus, he observes its accumulated mileage x_t and decides the replace decision, $d_t = 1$ if true and $d_t = 0$ otherwise.

The costs are as follows:

$$u_{t1} = -\beta^t(\theta_0 + \theta_1 x_t + \varepsilon_{t1})$$

$$u_{t2} = -\beta^t(\theta_1 x_t + \varepsilon_{t2})$$

Where ε_t is an iid Type I Extreme value random variable. The law of motion governing mileage is:

$$x_{t+1} = b_t + (1 - d_t)x_t$$

Where b_t is an iid random variable with discrete support $\{0, 1, 2\}$. For our commodity, we assign fixed probabilities for each scenario that do not depend



What now?

- Construct above the aforementioned repo and build the estimation routine.
- I'm meeting with my advisor every week on Monday or Tuesday.